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BREAST CANCER OBSCURED BY COSMETIC BREAST FILLER INJECTIONS: DIAGNOSTIC CHALLENGES AND IMAGING PITFALLS IN TWO CASES

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INTRODUCTION

Cosmetic breast filler injections may produce extensive inflammatory, fibrotic and nodular changes within the breast[1].

These changes can:

- Mimic malignant lesion
- Obscure underlying breast parenchyma
- Limit interpretation of conventional breast imaging
- Make identification of a reliable biopsy target challenging

CASE PRESENTATION

CASE 1 - 50 years old female, complained of left breast lump. History of filler injection in 2014.

With the given history of prior filler injection, she was scheduled for an MRI in a private center which demonstrated an enhancing lobulated lesion in the left lower inner quadrant (Figure 1a, asterisk). Subsequently, she underwent an excisional biopsy which revealed an invasive carcinoma with ductal carcinoma in situ.

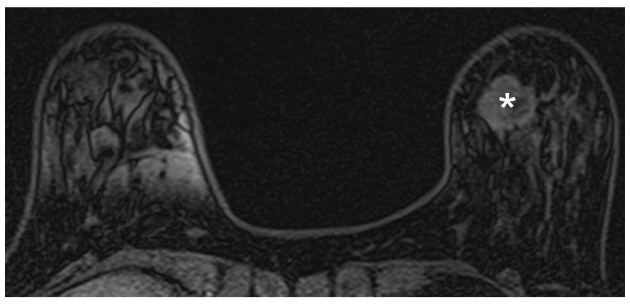

Figure 1a.

After the excisional biopsy, a lump was felt at the left breast and ultrasound examination was performed which demonstrated homogeneous hyperechoic dense shadowing throughout both breasts (Figure 1b) in keeping with filler related changes. Similar findings are also seen in the axilla likely filler infiltration into the lymph nodes (Figure 1c, asterisks).

Figure 1b.

Figure 1c.

CASE 2 - 60 years old female, complained of enlarging left breast lump.

Mammogram demonstrates diffuse diffuse high density opacities throughout both breasts, consistent with free silicone injections (Figure 2a). Complementary ultrasound shows high reflective material within both breast casting diffused posterior shadowing artefacts (snow storm appearance) with an ill-defined hypoechoic lesion identified at left 3 o'clock position (region of palpable hard lump)(Figure 2b). A contrast enhanced MRI breast was scheduled to further characterize the lesion. MRI demonstrates a spiculated heterogeneously enhancing solid mass in the left mid outer quadrant (Figure 2c, arrow). This lesion shows restricted diffusion and fast initial enhancement with plateau on the delayed phase (Type 2 kinetics). Image-guided biopsy on the same day confirmed an invasive carcinoma.

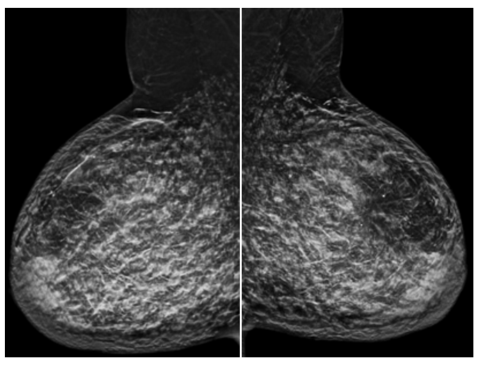

Figure 2a.

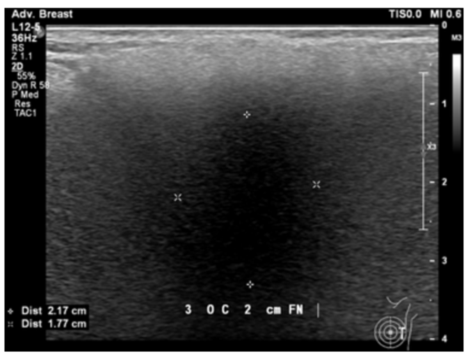

Figure 2b.

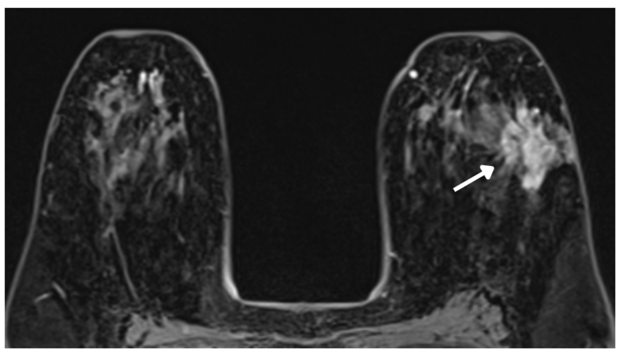

Figure 2c.

DISCUSSION

Breast filler injections can significantly compromise breast imaging by distorting normal architecture and producing appearances that mimic or obscure malignancy. Depending on the injected material, mammography may demonstrate dense opacities or calcifications, while ultrasound can be limited by marked posterior acoustic/reverberation ("snowstorm") artefacts[2]. Filler-related complications, including granulomatous inflammation, fat necrosis and fibrosis, may further mimic suspicious masses. The presence of breast fillers should therefore not lead to false reassurance or attribution of suspicious findings to filler-related changes alone. A detailed history of filler injection is important, while multimodality imaging, particularly contrast-enhanced MRI, may be required when conventional imaging findings are inconclusive[3]. Persistent or suspicious abnormalities warrant tissue diagnosis. This case highlights the importance of maintaining a high index of suspicion for co-existing breast malignancy despite the presence of cosmetic fillers[4].

CONCLUSION

A history of cosmetic breast filler injection should be actively sought in women undergoing breast assessment. Early escalation to contrast-enhanced imaging and a tailored biopsy strategy may reduce diagnostic delay and facilitate accurate staging and management when mammographic or sonographic evaluation is limited. Maintaining a high index of suspicion for malignancy remains essential, as filler-related changes may mask or mimic underlying breast cancer.

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DON'T LET THE FILLERS HIDE THE CANCER